KUKA

ROBOT

(V)KR 6; (V)KR 16; ...-2

Maintenance

In-line Wrist 6 kg II

Issued: 30 Sep 2009 **Version: 06**

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Other functions not described in this documentation may be operable in the controller. The user has no claim to these functions, however, in the case of a replacement or service work.
We have checked the content of this documentation for conformity with the hardware and software described. Nevertheless, discrepancies cannot be precluded, for which reason we are not able to guarantee total conformity. The information in this documentation is checked on a regular basis, however, and necessary corrections will be incorporated in subsequent editions.
Subject to technical alterations without an effect on the function.

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Valid for **all robots**
with in-line wrist 6 kg II

1 General



This description applies analogously to all the robots listed above, regardless of the robot variant or model shown in the illustrations.



The Doc. Module “Safety, General” is to be observed!

Before any maintenance work is started, any attached tools or additional equipment that would hinder maintenance must be dismounted.

Explanation of the symbols used in connection with maintenance intervals in the “Lubrication” and “Other maintenance” tables below:

Lubrication is performed either at the specified maintenance intervals or every 5 years after commissioning by the customer – whichever is reached first.

With maintenance intervals of 20,000 hours, the first maintenance (oil change) is performed either after 20,000 operating hours or 5 years after commissioning by the customer – whichever is reached first.



It is imperative that lubrication is carried out every 5 years, as prescribed. (Oil ageing!).



The maintenance intervals given in the tables are valid for the operating conditions specified in the robot Doc. Module “Technical Data”. In case of variations from normal conditions (e. g. increased dust or water content in the environment of the robot, abnormal temperatures), the KUKA design department must be consulted.

Explanation of symbols:



Maintenance point



Maintenance point, visible on drawing



Maintenance point, hidden on drawing



Oil change, to be repeated at specific intervals;

with indication of oil capacity



Lubricate with brush



Check toothed belt tension

The description of maintenance operations is subdivided into job steps with numbers in brackets appearing before them. The text which immediately follows these steps must also be read if it is specially marked by a **warning triangle** or either of the **hand symbols**. This is because many of these marked texts refer to the preceding job step.



Example:

(1) Pour specified amount of oil into the filler hole at (10).



Only KUKA-approved lubricants may be used.

Some of the specially marked texts refer exclusively to everything that follows – until the instruction is expressly revoked or the work is completed at the end of a section.



Example:



Turn main switch on the robot control cabinet to "OFF" and secure it with a padlock to prevent unauthorized persons from switching it on again.

2 Lubrication work



Chapter 1 of this Doc. Module is to be observed!



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.



Store used oil and grease in accordance with regulations and dispose of them with minimum environmental impact.

- **“Lubrication” table**

Maintenance point (see Fig. 1)	maintenance interval (hours) 10 000	20 000	Activity Lubricant Manufacturer Amount KUKA Art. no.	Comment
① Gear train A 4		●	Changing the oil Optigear Synthetic RO 150 Castrol, approx. 0.10 l [1)] KUKA Art. no. 00-144-898	Change oil as described in Section 2.1. Exchange 2.2.3.
② O-ring A 4	●		Lubricate Optigear Olit CLS Optimol, approx. 10 g per O-ring KUKA Art. no. 83-087-241	Lift O-rings out of grooves. Remove old grease. Grease O-rings with brush and fit them back into grooves.
③ Gear train A 5/6		●	Changing the oil Optigear Synthetic RO 150 Castrol, approx. 0.10 l [1)] KUKA Art. no. 00-144-898	Change oil as described in Section 2.1. Exchange 2.2.3.
④ O-ring A 5	●		Lubricate Optigear Olit CLS Optimol, approx. 10 g per O-ring KUKA Art. no. 83-087-241	Lift O-ring out of groove. Remove old grease. Grease O-ring with brush and fit it back into groove.
⑤ O-ring A 6	●		Lubricate Optigear Olit CLS Optimol, approx. 10 g per O-ring KUKA Art. no. 83-087-241	Lift O-ring out of groove. Remove old grease. Grease O-ring with brush and fit it back into groove.

1) The oil quantities specified are the actual amounts of oil in the gear unit at first filling.

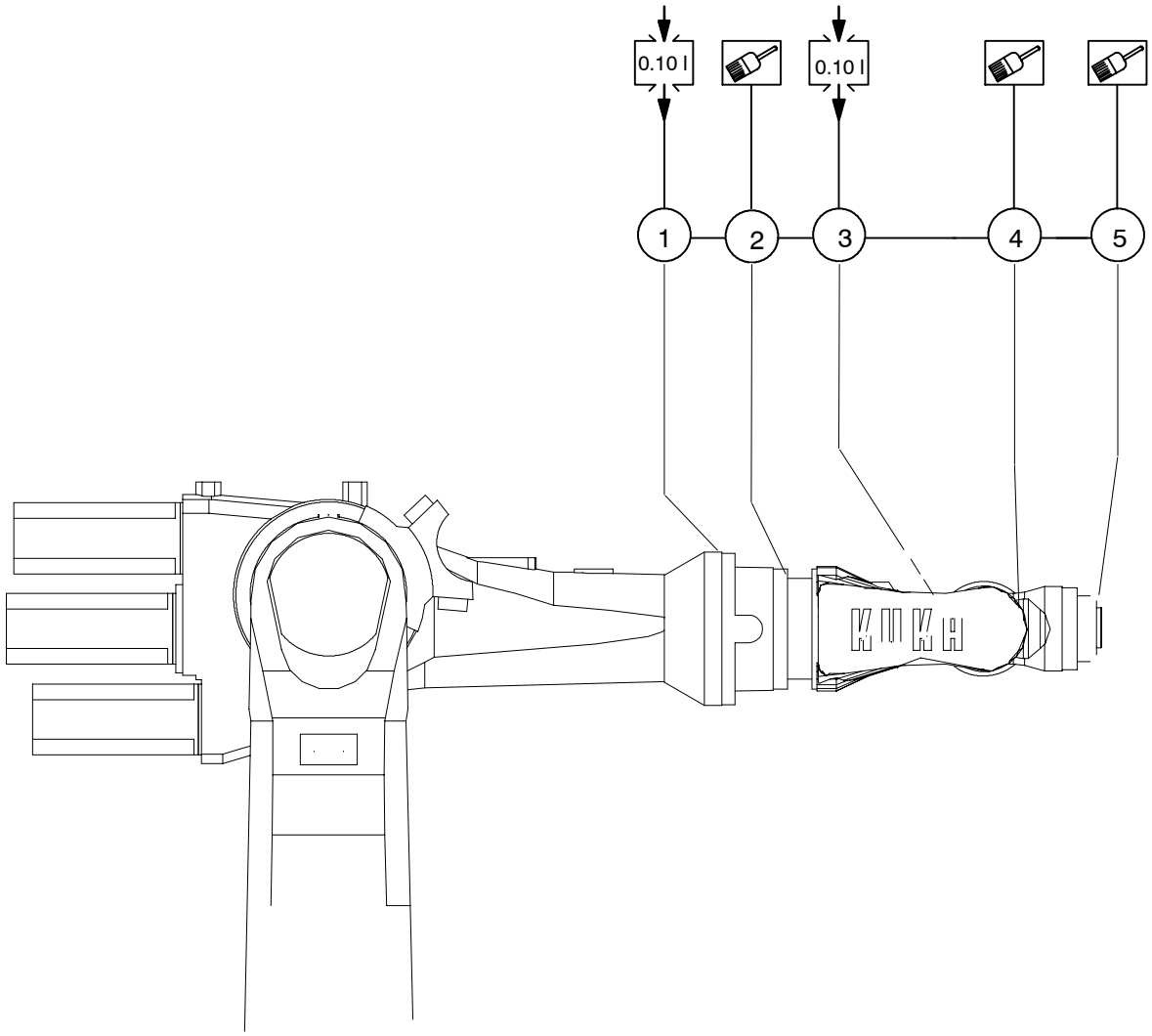


Fig. 1 Lubrication

When draining the oil, it is important to remember that the quantity drained is dependent on time and temperature.
The quantity of oil drained must be determined, as only this quantity may be used when refilling.

2.1 Oil change on in-line wrist



Chapter 2 of this Doc. Module is to be observed!



The oil is to be changed only at operating temperature.



If the oil change is carried out immediately after the robot has stopped operating, the oil temperature is liable to be high, in which case appropriate safety measures must be taken.

- **Drain oil**



Before performing the next step, it must be ensured that it is not possible for anyone to be injured within the range of the slowly rotating arm.
The robot may only be moved at jog speed, with all applicable safety rules and regulations being observed.

(1) Put robot into operation and move arm into a horizontal position (Fig. 2).

(2) Move wrist axes into the zero position.



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.

(3) Remove screw plugs (1, 2).

(4) Remove screw plugs (3, 4) and catch oil in a suitable container as it drains out.

(5) Check magnets of screw plugs for deposits, and clean magnets.



Store used oil in accordance with regulations and dispose of it with minimum environmental impact.

- **Fill with oil**

The arm is still positioned horizontally.

(1) If necessary, put robot into operation and move wrist axes into zero position.

(2) Renew aluminum sealing rings of the screw plugs.

(3) Screw in screw plugs (Fig. 2/3, 4) and pour specified amounts of oil into oil filler holes.

(4) Insert and tighten screw plugs (1, 2).
M10x1 screw plugs (1 – 4): tightening torque M_A = 10 Nm.

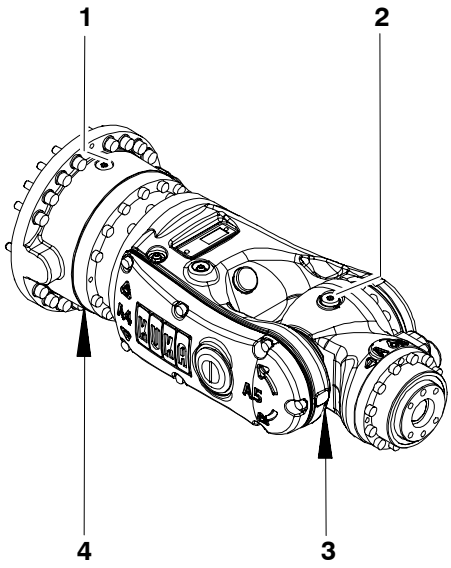


Fig. 2 Changing the oil

3 Other maintenance



Chapter 1 of this Doc. Module is to be observed!



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again.

Maintenance point (see Fig. 3)	Maintenance interval (hours) 5 000	10000	Activity	Comment
① Toothed belt In-line wrist 6 kg II A 5, A 6	● 1)		Check toothed belt tension	See robot Doc. Module “Adjustment Instructions”, KR 6, KR 16
			Check toothed belts	Replace toothed belts if cracks or signs of deterioration are found. See robot Doc. Module “Repair, In-line Wrist” KR 6, KR 16
		● 2)	Replace toothed belts	See robot Doc. Module “Repair, In-line Wrist” KR 6, KR 16

1) Carried out at KUKA before delivery, or must be performed by the user after a toothed belt has been exchanged.
2) After 2 years at the latest.

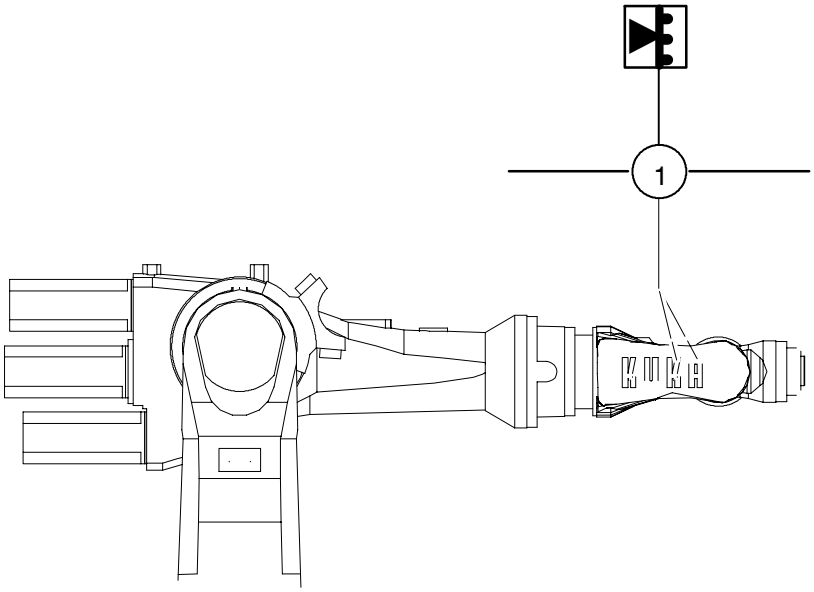


Fig. 3 Other maintenance

4 Cleaning and preventive maintenance



Turn main switch on the robot control cabinet to “OFF” and secure it with a padlock to prevent unauthorized persons from switching it on again. This also applies to any cleaning work, which though not performed directly on the robot is carried out within or close to its working range.

Cleaning and preventive maintenance work is to be carried out in accordance with the following instructions:

- Clean robot with a cloth soaked with washing agent.
- Clean cables, plastic parts and hoses with solvent-free washing agents.



The manufacturer’s instructions are to be observed when using washing agents.

- Remove any lubricant that has escaped using washing agent. If a large amount of lubricant has emerged, trace and remedy the cause.



It must be ensured that no washing fluid enters bearings, seals or electrical equipment.



If washing agent does enter bearings, these must be re-greased or re-oiled.

- Remove any corrosion and – where permissible – protect the affected areas with paint, grease or oil.
- Apply a thin coat of oil to bare metal parts.
- Replace damaged, illegible or missing inscriptions, labels and plates.



Do not use compressed air to clean the system, as this could cause dirt to penetrate and damage seals, bearings and electrical components.
Used cleaning agents and lubricants must be stored and disposed of in accordance with all pertinent regulations and with minimum environmental impact.

5 Safety precautions when handling lubricants

The information contained in the safety data sheet according to 91/155 EEC must be observed when handling lubricants. An excerpt from this data sheet is included in the robot Doc. Module “Consumables, Safety Data Sheet”.

The following safety procedures must be observed at all times:

- Avoid prolonged intensive skin contact; wear protective gloves and aprons if necessary (especially when working with cable grease).



All accident prevention regulations must be observed when handling lubricants.

- If lubricant is likely to come into contact with the skin, e.g. with the hands, suitable barrier creams must be applied before work is started.
- On completion of work and before smoking breaks or meal breaks (before any intake of food), clean oil-contaminated skin thoroughly with water and skin-friendly cleansing agents or soap. After cleaning, replace lost skin grease with appropriate skin creams containing grease.
- Change oil-soaked clothing immediately. Do not carry oily rags or any cloths soaked with solvents, mineral oil mixtures or cable grease in your pockets.
- Avoid breathing in oil spray or vapors wherever possible.